NHH

SCHOOL EXAMINATION

BAN404

Spring, 2025

Date: 02.06.2025

Time: 9.00-15.00

Number of hours: 6

An invigilator can contact course responsible by phone: 47078891

SUPPORT MATERIALS PERMITTED DURING THE EXAMINATION:

Calculator

All written support material permitted (category III)

Number of pages, including front page: 5

Number of attachments: 1, The file "data_task1.csv".

Instructions and recommendations

General recommendations: Be pragmatic if you run into difficulties. You need to find models that can answer the questions asked. Don´t let the perfect model stand in the way for a good one! Also, plan your work so that you do not use all time to solve one task.

R coding can be done in different ways, e.g. using base R or the tidyverse approach. Given a correct and clearly written result there is no preference for any particular approach.

Throughout the exam you will work with training and test data and you will have to make appropriate choices on when to use one or the other. In cases when you think there are more than one valid choice on working with the training, test or the full dataset, explain your choice!

Tasks

1. (50 points) This task is about some of the methodological topics treated in the course.

   (a) (10 points) Explain what the following R-functions are doing. Which method, treated in the course, is this a version of?

```r
# X is a n x p matrix with data
# y is an n x 1 vector with data

f <- function(b1,X,y,la)
{
  fval <- sum((y-X%*%b1)^2) + la*sum(b1^2)
  return(fval)
}

g <- function(X,y,la)
{
  q <- ncol(X)
  b0 <- mean(y)
  yd <- y - b0
  Xd <- X
  for(k in 1:q) Xd[,k] <- X[,k] - mean(X[,k])
  b1start <- rep(0,q)
  opt <- nlminb(b1start,f,X=Xd,y=yd,la=la)
  b1 <- opt$par
  return(b1)
}
```

   (b) (10 points) Consider the dataset in the file `data_task1.csv`. Extract the X- and y-values by

```r
# Read data
Xy <- read.csv2("data_task1.csv")
X <- as.matrix(Xy[,-1])
y <- as.matrix(Xy[,1])
```

   These are 100 observations of 20 independent variables and 1 dependent variable. Fit the model by ordinary least squares (OLS) (with the `lm`-function) and by using `g` with `la=1000`. Comment on the difference between the results. What is the explanation for the difference? Also, fit the model using `g` and let `la=0`.

   (c) (10 points) Explain what the following R-function does and use it to determine an optimal `la` for this dataset. Is it necessary to demean the x-variables? If so, why?

```r
loo <- function(la,X,y)
{
  n <- nrow(X)
  q <- ncol(X)
  b0 <- mean(y)
  yd <- y - b0
  Xd <- X
  for(k in 1:q) Xd[,k] <- X[,k] - mean(X[,k])
  ydpred <- matrix(0,n,1)
  for(i in 1:n)
  {
    b1 <- g(X=X[-i,],y=y[-i],la)
    ydpred[i] <- Xd[i,]%*%b1
  }
  testMSE <- mean((yd-ydpred)^2)
  return(testMSE)
}
```

(d) (10 points) Use the bootstrap method to estimate the sampling distribution of the sample variance of the variable `y` in `data_task1.csv`, $S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(y_i - \bar{y})^2$. Show the result as a histogram. Also, use the bootstrap to compute a 95% confidence interval of the variance of `y`, $\sigma_y^2 = \mathrm{Var}(y)$.

(e) (10 points) There might be one or more variables that have non-linear relationships to `y`. Try to identify such variable(s). Can you improve the in-sample (training) MSE by using a generalized additive model (GAM)? Compare the training MSE with at least one GAM. Also, use these MSE's to compute the share of explained variation, R^2, for the linear model estimated by OLS and for the GAM-model. If you did not obtain any MSE's from the models, or do not trust them, you can use $\mathrm{MSE}_{\mathrm{gam}} = 110000$ and $\mathrm{MSE}_{\mathrm{ols}} = 121000$.

2. (50 points) In this task, the dataset `dataOhlsson` in the R-package `insuranceData` shall be analyzed. Use `help(dataOhlsson)` to read about the dataset but note that the variable `kon` is gender and not age as said in the help-function.

(a) (10 points) Create a new variable called `claim` by recoding `antskad` to a binary variable (or a factor). Let it be equal to one ("yes") if `antskad`≥ 1 and zero ("no") otherwise.

```r
library(insuranceData)
data("dataOhlsson")
ydata=dataOhlsson
ydata$claim=ydata$antskad>=1
```

You are going to predict `claim` so remove variables that cannot plausibly be

assumed to be available when a prediction is made and explain your reasoning. Split the data into test and training data. Use descriptive statistics and graphs in order to find potentially useful predictors.

(b) (10 points) Fit a logistic regression model with all variables and one with a selection of variables. The selection can be based on your arguments in task 2a and by the result of the model with all variables.

(c) (10 points) Evaluate the predictions of the model in 2b on test data and with an appropriate (in)accuracy measure. Hint: You will first have to determine a threshold to convert from a probability to a prediction of the type `claim=0` or `claim=1`. You can experiment here and you do not need to do a formal cross-validation to find this threshold.

(d) (10 points) Fit a random forest and evaluate the predictions. Compare with the logistic model.

(e) (10 points) Fit a boosted tree and evaluate the predictions. Compare with the logistic model and the random forest.